

Programme	B. Sc. Computer Science				
Course Title	Virtualization				
Type of Course	Elective				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	11. Basic understanding of cloud computing				
Course Summary	This course introduces students to the fundamental concepts, technologies, virtualization, various virtualization tools and virtualization in storage, desktop, network and server				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand basics of virtualization	U	C	Instructor-created exams / Quiz
CO2	Understand how hypervisors work and their role in virtualization.	Ap	P	Assignment / Seminar presentations/ Exams
CO3	Understand Differences between various types of virtualization, including server virtualization, desktop virtualization, network virtualization, and storage virtualization	Ap	C	Seminar Presentation / Group Tutorial Work/ Viva Voce
CO4	Explore how virtualization technologies are used in the context of cloud services.	U	P	Instructor-created exams / Home Assignments
CO5	Understand the potential risks and vulnerabilities associated with virtualization and learn how to mitigate them.	U	P	Writing assignments/ Exams/ Seminar Presentations
CO6	Compare and analyse various virtualization tools	U	F	Case Study/ Exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48+12)	Marks (70)
I	Introduction to Virtualisation		12	18
	1	Virtualization and computing- need for virtualisation,	2	
	2	Cost, administration,	2	
	3	Fast deployment, reduce infrastructure cost	2	
	4	Limitations	1	
	5	Types of hardware virtualization: full virtualisation, partial virtualization, paravirtualization	3	
	6	Types of hypervisors	2	
II	Server and Desktop virtualization		14	20
	7	Virtual machine basics	2	
	8	Types of virtual machines	2	
	9	Understanding server virtualisation- types of server virtualization	3	
	10	Business cases for server virtualization	2	
	11	Uses of virtual server consolidation,	2	
	12	Selecting server virtualisation platform	1	
	13	Desktop virtualisation- types of desktop virtualization	2	
III	Network Virtualisation		12	16
	14	Introduction to network virtualisation	2	
	15	Advantages, functions	2	
	16	Tools for network virtualization	3	
	17	VLAN-WAN architecture	2	
	18	WAN Visualization	3	
IV	Storage Virtualization		10	16
	19	Introduction to memory virtualization	2	
	20	Types of storage virtualization	3	
	21	Risk of storage virtualization	2	
	22	SAN-NAS-RAID	3	
V	Open Ended Module- Virtualization tools (Any 3- \$ hours each)		12	
		VMWare-Amazon AWS Microsoft HyperV Oracle VM Virtual box IBM PowerVM Google Virtualization		

References

- Cloud Computing a practical approach- Anthony T Velte, Toby T Velte, Robert Elsenpeter, Tata McGraw Hill
- Virtualization from Desktop to the Enterprise, Chris Wolf, Eric M Halter

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	-	2	--	-	1	1						
CO 2	-	2	-	-	1	1						
CO 3	-	1	-	-	1	1						
CO 4	-	1	-	-	2	1						
CO 5	-	1	-	-	2	1						
CO 6	-	1	-	-	2	2						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	End Semester Examinations
CO 1	✓		✓
CO 2	✓		✓
CO 3	✓		✓
CO 4		✓	✓
CO 5		✓	✓
CO 6	✓	✓	